

Business Analytics Using Python

Final Report – Group 4

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Chapter 1

Introduction, Business Context, Dataset and Exploratory Analysis

1.1 Introduction and Business Context

Customer churn constitutes one of the most critical challenges faced by subscription-based service providers, particularly in highly competitive industries such as telecommunications. In mature markets where customer acquisition costs are substantial, retaining existing customers often represents a more economically efficient strategy than continuously attracting new ones. According to industry estimates, acquiring a new customer can cost between five and seven times more than retaining an existing one, making churn prevention a central strategic priority for telecom operators. Beyond direct revenue loss, churn also generates secondary negative effects, including deterioration of brand perception, reduced network effects, and increasing marketing expenditures. Consequently, modern telecommunications firms increasingly rely on data-driven approaches to identify customers at risk of churn and to understand the underlying drivers of defection. Predictive analytics offers a powerful framework for anticipating customer behavior, allowing firms to intervene proactively through targeted retention campaigns, personalized offers, or service improvements. This project focuses on the development of a predictive churn model using a publicly available telecommunications dataset. The objective is twofold. First, we aim to construct models capable of accurately identifying customers who are likely to churn. Second, and equally importantly, we seek to interpret the model outputs in order to understand why customers leave. This dual objective reflects a practical business constraint: a highly accurate model that lacks interpretability may be operationally limited, as decision-makers require clear explanations to design actionable strategies. The analysis therefore follows a progressive modeling approach. We begin with an interpretable linear model in the form of logistic regression, establishing a transparent baseline that directly quantifies the economic impact of key variables. Subsequently, we explore more complex

machine learning models—Random Forest and XGBoost—to enhance predictive performance. Finally, model-agnostic and model-specific explainability techniques, particularly SHAP values, are employed to reconcile predictive accuracy with interpretability. This methodological progression directly addresses the feedback received during the project review: rather than immediately relying on black-box models, we first extract insights from simpler statistical frameworks before leveraging more advanced algorithms.

1.2 Motivation and Conceptual Framework

Churn prediction is fundamentally a binary classification problem in which the outcome variable represents whether a customer terminates their service contract within a given observation period. However, unlike purely technical classification tasks, churn modeling is intrinsically tied to economic decision-making. The cost of false negatives (failing to identify a future churner) is typically much higher than that of false positives (incorrectly flagging a loyal customer), as missed churners translate directly into lost revenue. From a modeling perspective, this asymmetry motivates the use of probability-based outputs rather than simple class predictions. Instead of producing binary decisions alone, models generate churn probabilities, enabling business stakeholders to define operational thresholds that reflect marketing budgets, retention costs, and customer lifetime value. Furthermore, churn behavior is influenced by a mixture of demographic characteristics, contractual features, service usage patterns, and payment habits. Some drivers are structural, such as contract duration, while others are behavioral, such as billing preferences or technical support usage. Capturing these heterogeneous influences requires models capable of representing nonlinear relationships and feature interactions. At the same time, understanding the relative importance and direction of these drivers remains essential for strategic planning. For this reason, the project combines econometric modeling (logistic regression) with ensemble-based machine learning methods, supplemented by explainability frameworks.

1.3 Dataset Description and Preprocessing

The dataset used in this study contains 7,032 customer records after cleaning, each corresponding to an individual subscriber. The target variable, Churn, is binary, indicating whether the customer left the service (1) or remained (0). Approximately 26.6% of customers in the dataset have churned, revealing a moderately imbalanced class distribution that necessitates careful evaluation of recall and precision metrics in addition to overall accuracy. The explanatory variables cover four broad dimensions: customer demographics, account information, subscribed services, and billing characteristics. Demographic variables include gender and senior citizen status. Account-related variables include tenure

(length of customer relationship) and contract type. Service-related features capture access to internet services, online security, technical support, and streaming products. Financial variables include monthly charges, total charges, payment method, and paperless billing status. Before modeling, categorical variables were transformed into numerical representations. Binary variables such as Partner or PaperlessBilling were encoded as 0/1 indicators. Ordinal variables such as Contract were mapped to increasing numeric values reflecting commitment length (0 for month-to-month, 1 for one-year contracts, and 2 for two-year contracts). Certain service variables included intermediate values (0.5) to represent the absence of internet or phone services, allowing the model to distinguish between active non-subscription and structural unavailability. Continuous variables such as TotalCharges were converted to numeric format, and missing values were removed. For logistic regression, all selected features were standardized using z-score normalization to ensure coefficient comparability and numerical stability. The final feature set used across models includes tenure, MonthlyCharges, TotalCharges, Contract, OnlineSecurity, TechSupport, InternetService, PaymentMethod, PaperlessBilling, and SeniorCitizen. This selection reflects both domain relevance and empirical significance observed during preliminary analyses.

1.4 Exploratory Data Analysis

Initial descriptive statistics reveal several important structural properties of the dataset. Only 16.2% of customers are classified as senior citizens, indicating a demographic imbalance. However, subsequent analysis shows that senior status is associated with a higher likelihood of churn, suggesting that this smaller subgroup contributes disproportionately to customer attrition. Visualization of churn rates across demographic categories highlights notable patterns. Customers without partners or dependents exhibit higher churn proportions, suggesting that household structure may influence service stability. Similarly, customers on month-to-month contracts display significantly higher churn rates than those on longer commitments, an observation consistent with economic intuition. Financial variables also reveal strong associations with churn. Customers with higher monthly charges are more likely to leave, indicating potential price sensitivity or perceived lack of value. Conversely, longer-tenured customers exhibit markedly lower churn rates, reflecting relationship inertia and switching costs. Service-related variables provide additional insight. The absence of online security or technical support is correlated with increased churn, implying that customers who perceive insufficient service protection or assistance are more inclined to terminate their subscriptions. Internet service type also emerges as an important factor, with fiber-optic customers showing higher churn compared to DSL or non-internet users, possibly reflecting higher expectations associated with premium offerings. Correlation analysis confirms that no single variable fully ex-

plains churn, underscoring the multifactorial nature of customer defection. While tenure and contract length are strongly negatively correlated with churn, billing preferences and service attributes introduce additional layers of complexity. These exploratory findings motivate the adoption of multivariate modeling approaches capable of integrating diverse customer characteristics while controlling for confounding effects.

1.5 Transition Toward Predictive Modeling

Although exploratory analysis provides valuable descriptive insights, it remains insufficient for actionable decision-making. Individual variables exhibit moderate associations with churn, but business interventions require an integrated assessment of risk at the customer level. This necessitates predictive modeling frameworks that combine all relevant features into a unified probability estimate. In alignment with interpretability requirements, the analysis begins with logistic regression as a transparent baseline model. This approach enables direct economic interpretation through odds ratios while providing competitive predictive performance. Subsequently, Random Forest and XGBoost models are introduced to capture nonlinear relationships and higher-order interactions, with SHAP explainability techniques bridging the gap between accuracy and interpretability. The next section details this modeling pipeline, starting with logistic regression before progressing to ensemble methods.

Chapter 2

Methodology: From Interpretable Linear Models to Ensemble Learning

2.1 Modelling Strategy and Evaluation Framework

The modelling strategy adopted in this project follows a progressive approach, beginning with a transparent statistical model and subsequently introducing more complex machine learning algorithms. This sequence is intentional and aligns with the project's dual objectives: achieving strong predictive performance while maintaining interpretability. All models are trained using an 80/20 stratified train–test split to preserve the original churn distribution. Performance is evaluated using accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (ROC AUC). While accuracy provides a general measure of correctness, it is insufficient in isolation due to class imbalance. Therefore, particular emphasis is placed on recall for churners, as failing to identify customers who will leave represents the most costly error in business terms. Model outputs are treated as probabilities rather than direct class labels, enabling the application of adjustable decision thresholds to reflect operational constraints. This probabilistic framing is especially important for retention campaigns, where marketing resources are finite and interventions must be targeted toward customers with sufficiently high predicted risk. The modeling pipeline consists of three core components: logistic regression, Random Forest, and XGBoost. Each contributes distinct analytical value.

2.2 Logistic Regression as an Interpretable Baseline

Logistic regression is employed as the initial modeling framework due to its interpretability and well-established statistical properties. Unlike black-box algorithms, logistic regression directly estimates the relationship between explanatory variables and the log-odds of churn. This allows coefficients to be translated into odds ratios, offering immediate

economic meaning. Formally, the model estimates:

$$\log\left(\frac{(P = Churn)}{1 - (P = Churn)}\right) = \beta_0 + \sum_{i=1}^{k} \beta_i X_i \tag{2.1}$$

where (X_i) denotes standardized input features and (β_i) represents their associated coefficients. All continuous variables were standardized prior to estimation to ensure numerical stability and comparability across coefficients. The model was trained using maximum likelihood estimation, and convergence was achieved after eight iterations.

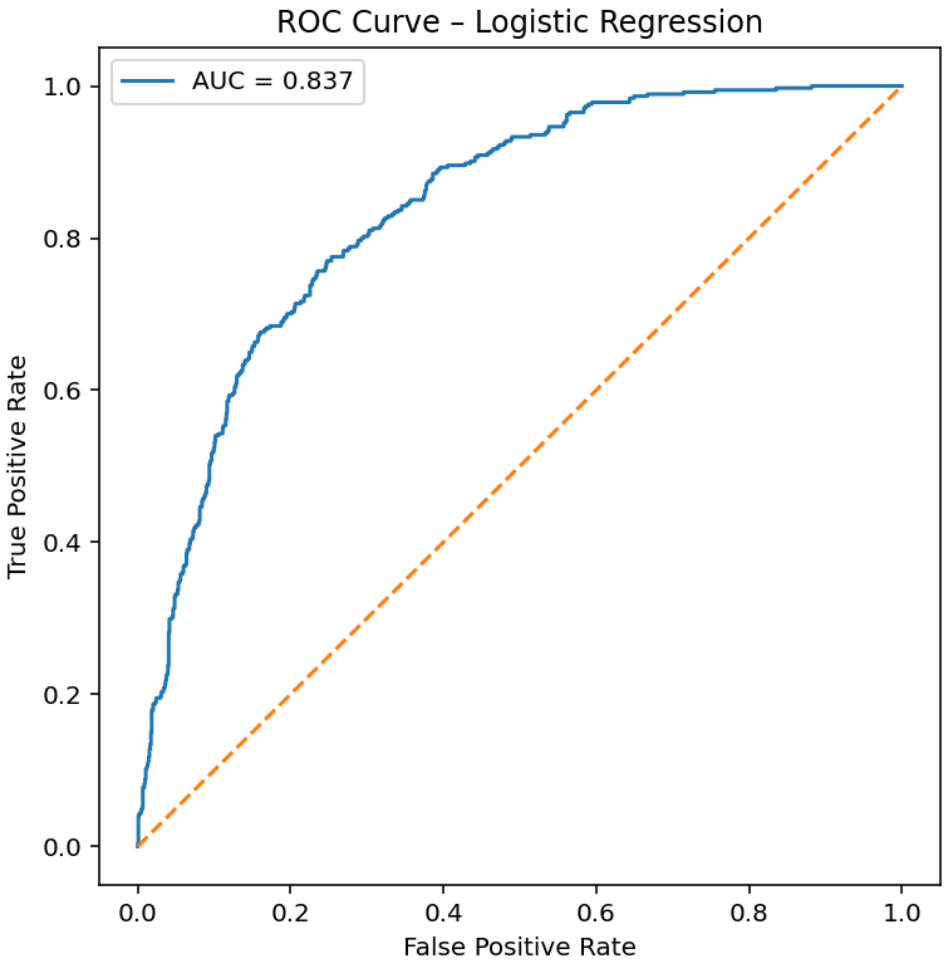


Figure 2.1: ROC Curve - Logistic Regression

The logistic regression model achieved an accuracy of approximately 80% and a ROC AUC of 0.84, indicating strong discriminative power despite its simplicity. Precision for churners reached 0.64, while recall was 0.57, reflecting a reasonable but improvable ability to capture customers at risk.

Beyond predictive performance, the primary strength of logistic regression lies in its interpretability. Odds ratios derived from the fitted coefficients provide direct insight into how each feature influences churn probability. Tenure exhibits an odds ratio of approximately 0.24, making it the strongest protective factor in the model. This implies that

longer-tenured customers are dramatically less likely to churn, consistent with the presence of switching costs and relationship inertia. Each standard deviation increase in tenure reduces churn odds by roughly 76%, highlighting customer longevity as a key stabilizing force. Contract length also shows a substantial protective effect, with an odds ratio near 0.57. This indicates that moving from month-to-month to longer contracts significantly reduces churn risk. Economically, this reflects commitment effects: customers locked into longer agreements face higher exit barriers and are less responsive to short-term dissatisfaction. Conversely, TotalCharges displays an odds ratio above 2, suggesting that higher cumulative spending is associated with increased churn risk. This counterintuitive result may reflect dissatisfaction among high-paying customers who perceive insufficient value for money, or it may capture customers nearing contract renewal periods after extended usage. InternetService emerges as a strong positive driver of churn, with an odds ratio of approximately 1.58. This suggests that customers subscribed to fiber-optic or advanced internet services are more likely to leave, potentially due to heightened expectations or aggressive competition in premium segments. PaperlessBilling and SeniorCitizen status also increase churn odds, though to a lesser extent. The elevated churn among seniors may reflect lower technological comfort or greater sensitivity to pricing and service complexity. Service-related variables such as OnlineSecurity and TechSupport exhibit odds ratios below one, indicating protective effects. Customers benefiting from these services appear more satisfied or more embedded in the provider's ecosystem, reducing their likelihood of defection. Importantly, several variables demonstrate statistical significance at conventional levels, confirming that their effects are unlikely to be due to random variation. Together, these results provide a coherent economic narrative: churn is primarily driven by contract flexibility, service quality, pricing perception, and customer tenure. This interpretable baseline establishes a foundation for understanding churn behavior before introducing more complex models.

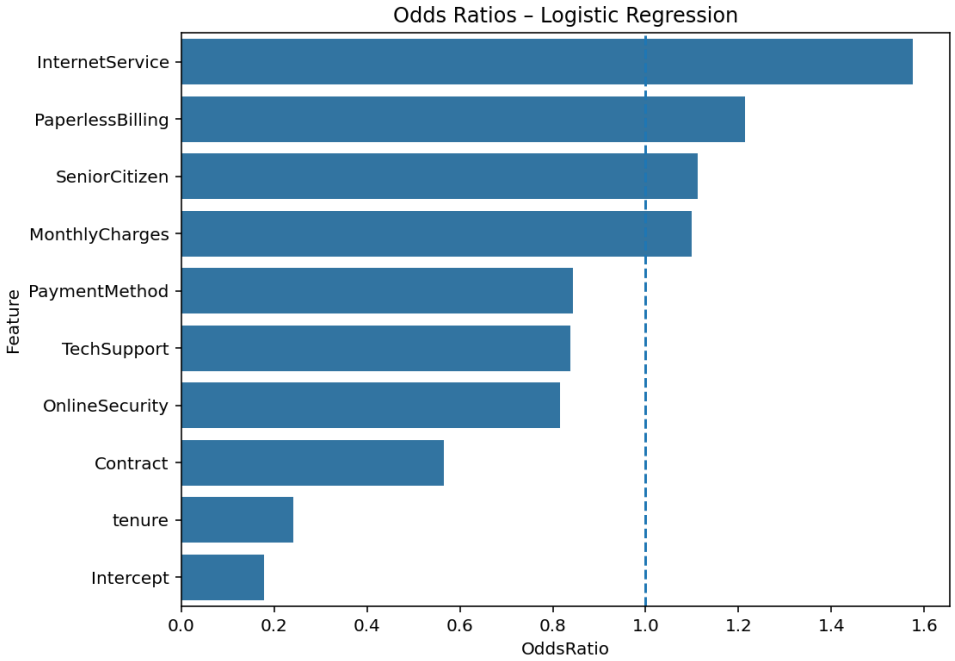


Figure 2.2: Odds Ratios

2.3 Random Forest Classification

While logistic regression offers clarity, it imposes linearity assumptions and cannot naturally capture complex feature interactions. To overcome these limitations, a Random Forest classifier is introduced as the first nonlinear machine learning model. Random Forest operates by constructing an ensemble of decision trees trained on bootstrapped samples of the data, with random subsets of features considered at each split. This process reduces variance relative to individual trees and allows the model to capture nonlinear patterns without extensive manual feature engineering. In this project, the Random Forest model was trained using 300 trees with class balancing to address churn imbalance. The model achieved an accuracy of approximately 74% and a ROC AUC of around 0.80. Although slightly inferior to logistic regression in terms of AUC, the Random Forest model offers complementary insights through feature importance measures and nonlinear decision boundaries. Feature importance analysis indicates that contract length, internet service, and billing-related variables play dominant roles in prediction. However, traditional Random Forest importance scores suffer from bias toward continuous variables and do not provide directional information, motivating the use of SHAP explainability in later stages. Random Forest's main contribution lies in validating the nonlinear nature of churn drivers and identifying interaction effects that linear models may overlook. However, its limited interpretability relative to logistic regression constrains its standalone business applicability.

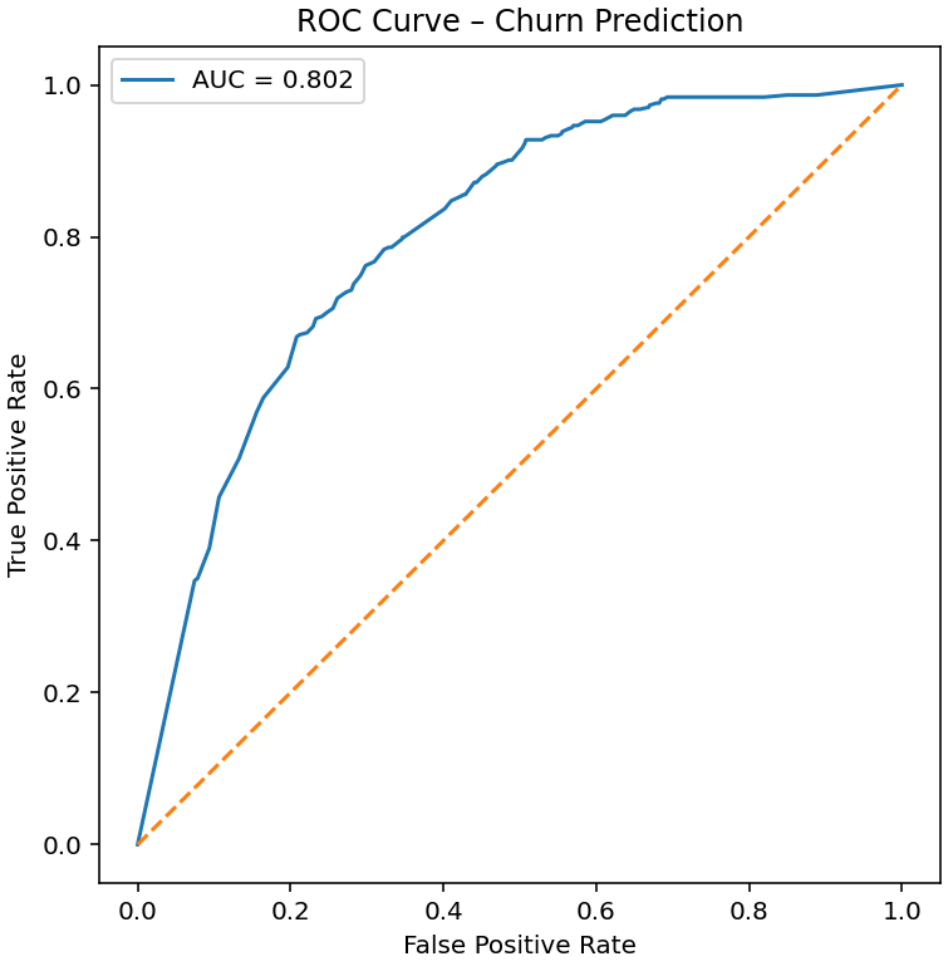


Figure 2.3: ROC Curve - RandomForestClassifier

2.4 Gradient Boosting with XGBoost

To further enhance predictive performance, the project employs Extreme Gradient Boosting (XGBoost), a state-of-the-art ensemble method based on sequential tree boosting. Unlike Random Forest, which builds trees independently, XGBoost constructs trees iteratively, with each new tree correcting the errors of its predecessors. The XGBoost model was configured with 400 estimators, moderate tree depth, subsampling, and class imbalance adjustments. This architecture balances model complexity and generalization. XGBoost achieved a ROC AUC exceeding 0.83, outperforming Random Forest and matching logistic regression while offering superior flexibility. By tuning the probability threshold to 0.35, recall for churners was increased, reflecting a strategic emphasis on identifying at-risk customers even at the cost of additional false positives. This threshold adjustment demonstrates the practical importance of probability-based outputs. Rather than adhering to the default 0.5 cutoff, the decision boundary is shifted to align with business objectives, prioritizing customer retention over classification purity. XGBoost

also provides built-in feature importance metrics, but, as with Random Forest, these lack directional clarity. Consequently, SHAP values are employed to obtain both global and local explanations of model behavior.

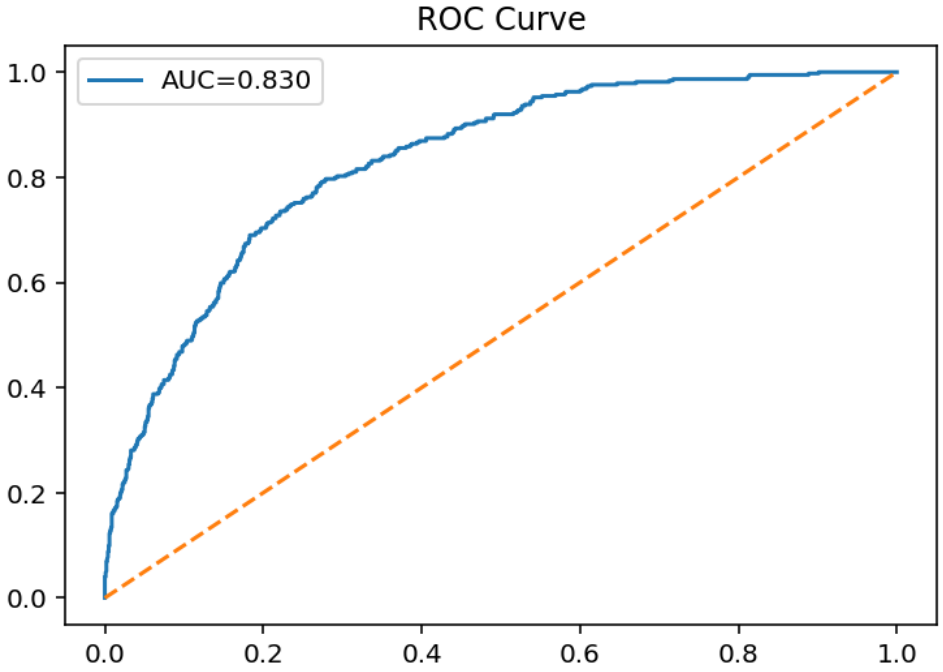


Figure 2.4: ROC Curve - XGBoost

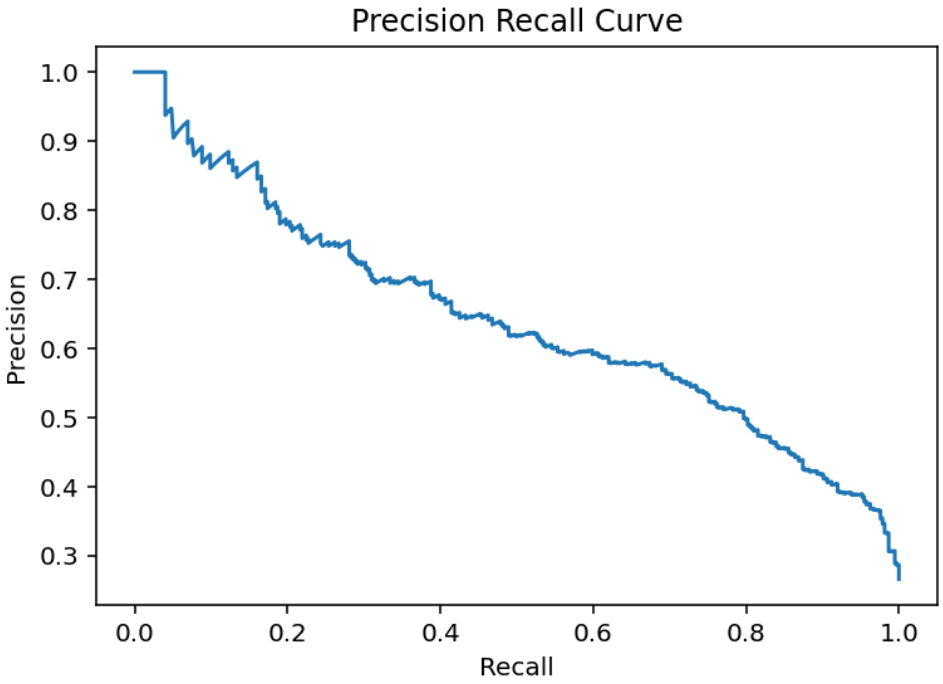


Figure 2.5: Precision / Recall - XGBoost

2.5 Summary of Methodological Progression

The modelling pipeline intentionally progresses from interpretable linear regression to increasingly sophisticated machine learning algorithms. Logistic regression establishes causal-like associations and economic interpretability. Random Forest introduces nonlinear structure and interaction effects. XGBoost maximizes predictive performance while retaining explainability through SHAP. This layered approach reconciles two often competing goals in applied analytics: understanding customer behavior and accurately predicting outcomes. Rather than treating black-box models as replacements for traditional statistics, they are positioned as complementary tools, enriching insights derived from simpler frameworks. The following section presents a detailed analysis of model performance, explainability results, and business implications.

Chapter 3

Model Evaluation, Explainability, and Interpretation of Churn Drivers

3.1 Evaluation Metrics and Predictive Performance

In churn prediction problems, traditional accuracy metrics provide an incomplete picture of model effectiveness. Because the dataset exhibits class imbalance, with churners representing roughly 26% of observations, a naïve classifier that predicts "no churn" for every customer would already achieve an accuracy close to 74%. Consequently, the evaluation framework emphasizes probabilistic discrimination and class-specific performance rather than raw accuracy alone. The primary metric used throughout this project is the area under the Receiver Operating Characteristic curve (ROC AUC). ROC AUC measures the probability that the model assigns a higher churn probability to a randomly selected churner than to a randomly selected non-churner. Unlike accuracy, it is insensitive to class imbalance and captures the ranking quality of predicted probabilities across all possible thresholds. The logistic regression model achieved a ROC AUC of approximately 0.84, demonstrating strong discriminatory power despite its linear structure. Random Forest reached a ROC AUC close to 0.77, while XGBoost achieved values exceeding 0.83 after hyperparameter tuning. These results illustrate that even relatively simple linear models can perform competitively when features are carefully selected and standardized, while gradient boosting provides incremental gains through nonlinear modeling. Precision and recall were also examined to assess the trade-off between identifying churners and avoiding false positives. Logistic regression yielded a recall of approximately 57% for churners, meaning that slightly more than half of customers who eventually churn are correctly detected. Precision for churners reached 64%, indicating that nearly two-thirds of customers flagged as high-risk truly churn. Random Forest exhibited lower recall, reflecting its more conservative predictions. XGBoost, by contrast, allowed explicit adjustment of the decision threshold to prioritize recall. By lowering the probability cutoff from the

default 0.5 to 0.35, recall increased substantially, at the expense of reduced precision. This threshold tuning highlights an essential operational insight: churn prediction is not a purely statistical task but a business decision problem. A telecom operator may prefer to contact more customers preemptively, even if some would not have churned, provided the cost of intervention is outweighed by retained revenue. Thus, probability thresholds must be chosen in alignment with marketing budgets and customer lifetime value rather than arbitrary classification conventions.

3.2 Probability Thresholding and Business Optimization

All three models generate continuous churn probabilities. These probabilities represent estimated individual risk levels and form the basis for actionable retention strategies. Rather than classifying customers using a fixed 0.5 threshold, the project explicitly treats churn prediction as a ranking problem. Customers are ordered by predicted churn probability, and interventions are targeted toward those exceeding a chosen cutoff. In the XGBoost model, a threshold of 0.35 was adopted to increase sensitivity toward churners. This adjustment increased recall, enabling the identification of a larger proportion of customers at risk, which is particularly valuable in retention contexts where missing a potential churner carries significant financial consequences. From a business perspective, this approach allows management to simulate different operational scenarios. A high threshold focuses on the most critical cases but misses many moderate-risk customers. A lower threshold captures more churners but increases outreach costs. The optimal threshold depends on campaign capacity, average customer lifetime value, and intervention effectiveness. This probabilistic framework transforms machine learning outputs into a decision-support tool rather than a rigid classifier.

3.3 Explainability Through Odds Ratios and SHAP Values

While performance metrics quantify predictive accuracy, they do not explain why customers churn. Understanding drivers is essential for designing targeted interventions and informing strategic decisions. This project therefore combines two complementary explainability approaches: odds ratios derived from logistic regression and SHAP values applied to ensemble models.

3.3.1 Logistic Regression Odds Ratios

The logistic regression model provides direct interpretability via odds ratios. These quantify how a one-standard-deviation increase in a feature affects the odds of churn, holding other variables constant. Tenure emerges as the dominant protective factor, with an odds ratio near 0.24. This implies that customers with longer tenure are dramatically less likely to churn, reflecting accumulated switching costs, familiarity with services, and relational inertia. Contract length also exhibits a strong protective effect, with an odds ratio of approximately 0.57. Moving from month-to-month contracts toward longer commitments substantially reduces churn probability. This result highlights contractual structure as a central lever in churn prevention. Service-related variables such as OnlineSecurity and TechSupport further decrease churn odds, suggesting that customers receiving support and security services feel more satisfied or embedded in the provider ecosystem. Conversely, TotalCharges displays an odds ratio above 2, indicating that customers who have paid more cumulatively are more likely to churn. This counterintuitive finding may reflect heightened expectations among high-paying customers or dissatisfaction with perceived value. It may also capture customers approaching contract renewal after extended usage periods. InternetService shows a positive association with churn, implying higher churn among customers using advanced connectivity solutions. This likely reflects competitive pressure in premium segments, where customers have more alternatives and higher service expectations. SeniorCitizen status modestly increases churn odds, potentially reflecting usability challenges or greater price sensitivity among older customers. Together, these coefficients form a coherent economic narrative: churn is driven primarily by contract flexibility, tenure, service quality, and perceived value.

3.3.2 SHAP Explainability for Tree-Based Models

While logistic regression provides global linear effects, it cannot capture complex interactions. To interpret Random Forest and XGBoost models, SHAP (SHapley Additive exPlanations) values are employed. SHAP decomposes each individual prediction into additive feature contributions, grounded in cooperative game theory. For each customer, SHAP quantifies how much each variable pushes the prediction toward churn or retention relative to a baseline. Global SHAP summary plots reveal that tenure, contract type, internet service, TotalCharges, and MonthlyCharges consistently rank among the most influential features. This confirms the robustness of insights obtained from logistic regression while extending them to nonlinear contexts. Crucially, SHAP provides directional information. For example, low tenure values strongly increase churn probability, while long tenure reduces it. Month-to-month contracts push predictions toward churn, whereas long-term contracts pull them toward retention. High TotalCharges contribute positively to churn risk, while TechSupport and OnlineSecurity reduce it. Beyond global

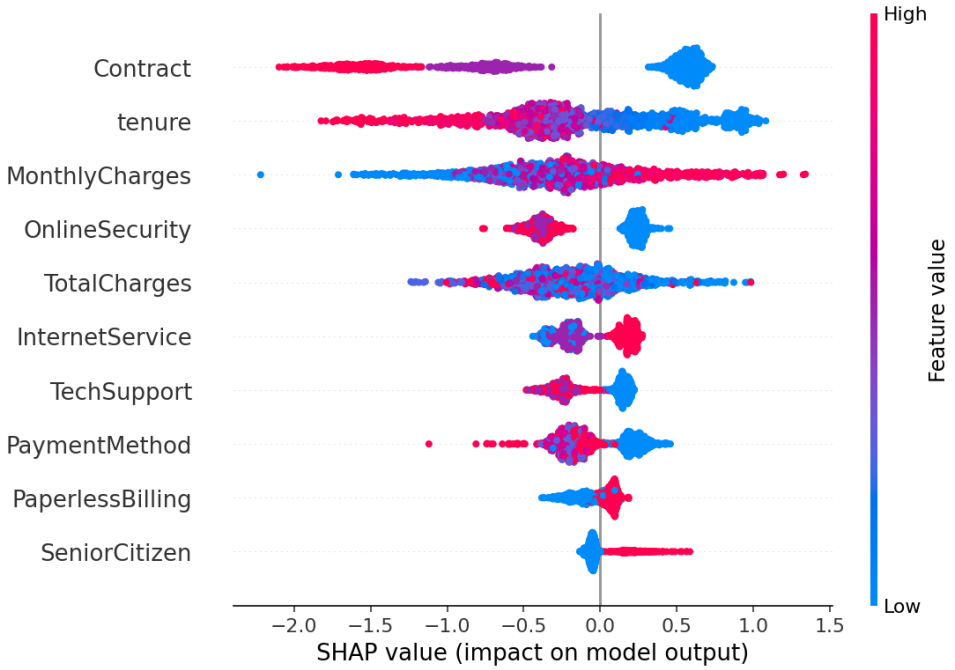


Figure 3.1: SHAP Explainability

patterns, SHAP enables local explainability. For any individual customer, it becomes possible to explain precisely why the model predicts high churn risk. A given customer may be flagged due to short tenure, absence of tech support, and high monthly charges, while another may be at risk primarily because of contract type and payment method. This individual-level transparency is essential for operational deployment. Retention teams can tailor interventions based on each customer's specific risk profile rather than applying generic offers.

3.4 Comparison of Logistic Regression, Random Forest, and XGBoost

Each modelling approach contributes distinct value to the analysis. Logistic regression offers clarity and economic interpretability. Its coefficients translate directly into actionable insights, making it ideal for communicating results to non-technical stakeholders. It also achieves strong ROC AUC, demonstrating that interpretability need not come at the expense of performance. Random Forest introduces nonlinear modelling and interaction effects but provides limited interpretability through traditional feature importance. Its performance is slightly weaker than logistic regression and XGBoost in this application, suggesting that the underlying relationships may be sufficiently captured by simpler structures. XGBoost delivers the strongest predictive flexibility and enables probability threshold optimization. When combined with SHAP, it offers both high performance and

rich explainability, making it suitable for operational deployment. Rather than competing approaches, these models form a complementary toolkit. Logistic regression establishes causal-like associations and validates feature relevance. XGBoost refines predictions and supports individualized risk assessment. SHAP bridges the interpretability gap between linear and nonlinear models.

3.5 Identification of High-Risk Customers

Using XGBoost probabilities, customers in the test set are ranked by churn risk. This ranking enables the creation of a prioritized list of individuals for retention campaigns. Customers at the top of this list typically exhibit short tenure, month-to-month contracts, absence of technical support, and high charges. This ranking approach aligns with practical business workflows, where marketing teams must decide whom to contact under budget constraints. Instead of treating churn prediction as a binary classification problem, the model supports continuous prioritization based on estimated risk.

3.6 Synthesis of Findings

Across all models and interpretability frameworks, a highly consistent pattern emerges : customer churn is a multifactorial phenomenon driven by contractual structure, lifecycle dynamics, service quality, and pricing perception. First, tenure represents the most decisive stabilizing factor. Both logistic regression and SHAP analysis confirm a strong negative relationship between tenure and churn, indicating that early-stage customers are the most vulnerable segment. Second, contract type is the dominant structural driver of churn. Feature importance and SHAP values consistently rank contract length as the most influential predictor, demonstrating that flexibility significantly increases switching behavior. Third, pricing variables (MonthlyCharges and TotalCharges) exhibit positive associations with churn, suggesting that perceived value plays a central role in customer retention. High-paying customers appear more sensitive to service quality and competitive alternatives. Fourth, service quality variables such as OnlineSecurity and TechSupport act as protective factors, reinforcing the hypothesis that ecosystem integration and perceived support reduce defection risk. The convergence of results across logistic regression, Random Forest, XGBoost, and SHAP explainability significantly strengthens the internal validity of the study and demonstrates the robustness of the identified churn drivers.

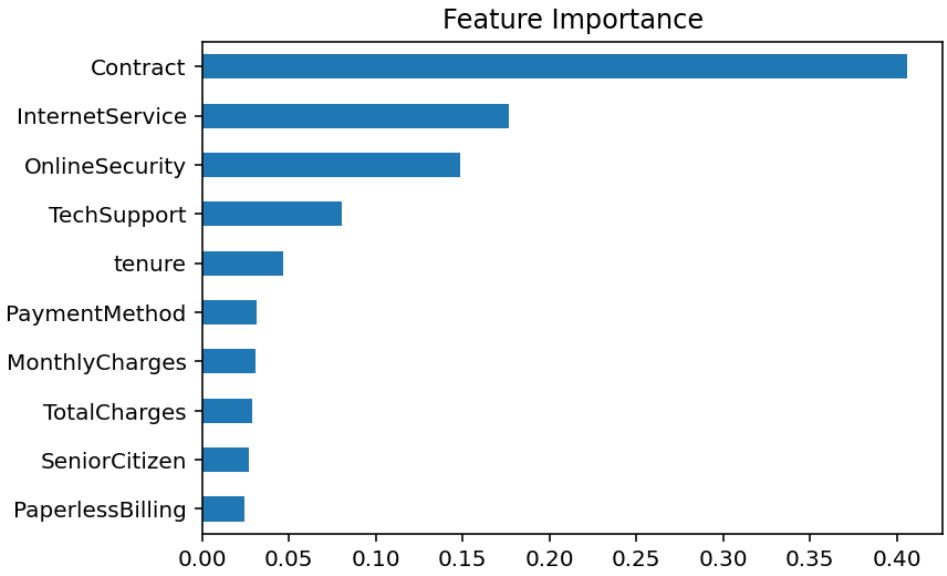


Figure 3.2: 10 Most Important Features

Chapter 4

Business Implications, Limitations, Future Work, and Conclusion

4.1 Translating Analytical Results into Business Strategy

The analytical results provide a clear foundation for data-driven retention strategies. The most critical managerial insight concerns customer lifecycle management. Since low-tenure customers exhibit the highest churn probability, telecom firms should prioritize onboarding optimization through early engagement campaigns, personalized guidance, and proactive support during the first months of subscription. Contract restructuring represents a second strategic lever. Encouraging migration from month-to-month contracts to annual plans through discounts, loyalty rewards, or bundled services could significantly reduce churn risk. Additionally, the protective effect of OnlineSecurity and TechSupport suggests that bundling support services with premium plans could increase perceived value and customer lock-in. From a segmentation perspective, the model identifies three priority risk profiles:

- High-risk: Low tenure + Month-to-month contract + High charges
- Medium-risk: Digital billing users with limited support services
- Low-risk: Long-tenure customers with bundled services and long contracts

This segmentation enables targeted retention campaigns rather than costly mass marketing interventions, improving both efficiency and customer lifetime value.

4.2 Operational Deployment and Decision Framework

From an operational standpoint, the model outputs should be viewed as probabilistic risk scores rather than binary classifications. The use of adjustable probability thresholds allows managers to balance recall and precision depending on business objectives and re-

source constraints. For example, a low threshold can be used during high-value campaigns to capture as many churners as possible, accepting higher false-positive rates. Conversely, a higher threshold may be preferable when outreach capacity is limited or intervention costs are high. A practical deployment pipeline would involve regularly scoring the active customer base, ranking customers by churn probability, and selecting the top segment for retention actions. SHAP explanations can be attached to each customer record to guide personalized interventions, enabling call center agents or automated systems to address the most relevant issues for each individual. Such a system transforms churn prediction into a continuous decision-support process integrated within customer relationship management workflows.

4.3 Addressing the Professor's Feedback

The professor's critique emphasized two key points: the initial exploratory analysis appeared disconnected from the modelling stage, and the project moved too quickly toward black-box models without sufficient emphasis on interpretability. This feedback has been explicitly addressed in the final analytical pipeline. First, logistic regression was introduced as a baseline explainable model. This step reconnects the exploratory phase with predictive modelling by quantifying relationships observed in descriptive statistics and correlation analysis. Odds ratios provide direct economic interpretation and validate the relevance of key variables such as tenure, contract type, and service features. Second, more complex models (Random Forest and XGBoost) were introduced only after establishing this interpretable foundation. Their role is not to replace linear modelling but to refine predictions and capture nonlinear interactions. SHAP explainability bridges the gap between performance and interpretability, ensuring that black-box models remain transparent. This sequential approach—from exploratory analysis to logistic regression to tree-based models—aligns with best practices in applied analytics and ensures that predictive accuracy does not come at the expense of understanding.

4.4 Limitations of the Study

Despite its strengths, this project has several limitations. First, the analysis relies on a static snapshot of customer data. Churn is inherently dynamic, and temporal patterns such as changes in usage behavior or billing trends are not captured. Incorporating time-series features could significantly enhance predictive power. Second, categorical variables were encoded using ordinal mappings (for example, contract type and payment method). While practical, this approach imposes artificial ordering and may distort relationships. One-hot encoding or target encoding could provide more faithful representations. Third, the models focus exclusively on correlation rather than causation. While logistic regression

coefficients and SHAP values identify associations, they do not establish causal effects. For example, offering TechSupport may reduce churn, but customers who choose TechSupport may already be more loyal. Without experimental or quasi-experimental designs, causal claims remain speculative. Fourth, the dataset lacks behavioral variables such as call frequency, complaint history, or service outages. These operational metrics are often strong predictors of churn and would likely improve model performance if available. Finally, the evaluation is conducted on a single train-test split. Although stratification mitigates sampling bias, cross-validation would provide more robust performance estimates.

4.5 Potential Extensions and Future Work

Several avenues could extend this work. Unsupervised clustering could be applied to segment customers based on behavioral and demographic profiles, providing complementary insights into heterogeneous churn patterns. Such clusters could inform differentiated marketing strategies. Uplift modelling represents another promising direction. Rather than predicting who will churn, uplift models estimate who will respond positively to retention interventions. This distinction is crucial, as not all high-risk customers are equally persuadable. Causal inference techniques, such as propensity score matching or instrumental variables, could help isolate the true impact of services like TechSupport or contract changes on churn. Deep learning models, particularly recurrent neural networks, could capture temporal dynamics if longitudinal data were available. Finally, integrating economic metrics such as customer lifetime value would allow optimization of retention strategies based on expected financial returns rather than churn probability alone.

4.6 Conclusion

This project demonstrates how a structured and explainable machine learning pipeline can transform raw customer data into actionable business intelligence. By progressively integrating logistic regression, Random Forest, and XGBoost, the analysis successfully reconciles predictive performance with managerial interpretability. The results consistently identify tenure, contract structure, service quality, and pricing perception as the core determinants of customer churn. Logistic regression provides transparent economic interpretation through odds ratios, while SHAP explainability extends interpretability to nonlinear ensemble models. Importantly, the study highlights that churn prediction is not merely a technical exercise but a strategic decision-support framework. Probability-based risk scoring, threshold optimization, and individualized SHAP explanations enable telecom operators to implement targeted, cost-efficient retention strategies. The project establishes a clear methodological progression from exploratory analysis to interpretable modelling and finally to advanced machine learning with explainability. This approach

ensures that predictive accuracy is complemented by analytical transparency and business relevance. Overall, the proposed framework provides a scalable and operationally deployable solution for churn management, with strong potential to enhance customer retention, optimize marketing resource allocation, and increase long-term customer lifetime value.

Appendix A

Appendices

A.1 Logistic Regression

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, accuracy_score,
    roc_auc_score, roc_curve

import statsmodels.api as sm

# Data Loading

churn_csv = r"C:\Users\mehdi\Documents\Mehdi\HEC\Cours\Python\Project\
    Telco_Customer_Churn.csv"

df = pd.read_csv(churn_csv)

df['Churn'] = df['Churn'].map({'Yes':1,'No':0})
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df = df.dropna()

df = df.drop(columns=['customerID'])

binary = {'Yes':1,'No':0}

df['gender'] = df['gender'].map({'Female':1,'Male':0})
df['Partner'] = df['Partner'].map(binary)
df['Dependents'] = df['Dependents'].map(binary)
df['PhoneService'] = df['PhoneService'].map(binary)
```

```
df['PaperlessBilling'] = df['PaperlessBilling'].map(binary)

service = {'Yes':1,'No':0,'No internet service':0.5}

df['OnlineSecurity'] = df['OnlineSecurity'].map(service)
df['OnlineBackup'] = df['OnlineBackup'].map(service)
df['DeviceProtection'] = df['DeviceProtection'].map(service)
df['TechSupport'] = df['TechSupport'].map(service)
df['StreamingTV'] = df['StreamingTV'].map(service)
df['StreamingMovies'] = df['StreamingMovies'].map(service)

df['MultipleLines'] = df['MultipleLines'].map({'Yes':1,'No':0,'No phone
    service':0.5})
df['InternetService'] = df['InternetService'].map({'Fiber optic':1,'DSL'
    :0.5,'No':0})
df['Contract'] = df['Contract'].map({'Month-to-month':0,'One year':1,'
    Two year':2})

df['PaymentMethod'] = df['PaymentMethod'].map({
    'Electronic check':0,
    'Mailed check':0.3,
    'Bank transfer (automatic)':0.7,
    'Credit card (automatic)':1
})

# Features

features = [
    'tenure','MonthlyCharges','TotalCharges',
    'Contract','OnlineSecurity','TechSupport',
    'InternetService','PaymentMethod',
    'PaperlessBilling','SeniorCitizen'
]

X = df[features]
y = df['Churn']

# Standardization

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Train Test Split

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y,
    test_size=0.2,
```

```
    stratify=y,
    random_state=42
)

# Logistic Regression

lr = LogisticRegression(max_iter=2000)
lr.fit(X_train, y_train)

y_prob = lr.predict_proba(X_test)[:,1]
y_pred = lr.predict(X_test)

print("\nLOGISTIC REGRESSION PERFORMANCE\n")
print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
auc = roc_auc_score(y_test, y_prob)
print("ROC AUC:", auc)

# ROC Curve

fpr, tpr, _ = roc_curve(y_test, y_prob)

plt.figure(figsize=(6,6))
plt.plot(fpr, tpr, label=f"AUC = {auc:.3f}")
plt.plot([0,1],[0,1],'--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - Logistic Regression")
plt.legend()
plt.show()

# Economic Interpretation

X_sm = sm.add_constant(X_scaled)

logit = sm.Logit(y, X_sm)
result = logit.fit()

print(result.summary())

# Odds Ratios

odds = pd.DataFrame({
    "Feature": ["Intercept"] + features,
    "OddsRatio": np.exp(result.params)
}).sort_values("OddsRatio", ascending=False)

```

```
print("\nODDS RATIOS\n")
print(odds)

# Visualization Odds Ratios

plt.figure(figsize=(8,6))
sns.barplot(data=odds[1:], x="OddsRatio", y="Feature")
plt.title("Odds Ratios - Logistic Regression")
plt.axvline(1, linestyle="--")
plt.show()
```

A.2 Random Forest Classifier

```
133 from sklearn.ensemble import RandomForestClassifier
134 from sklearn.model_selection import train_test_split
135 from mlxtend.feature_selection import SequentialFeatureSelector as SFS
136 from sklearn.metrics import accuracy_score, classification_report,
       recall_score, roc_auc_score, roc_curve, precision_recall_curve
137 import seaborn as sns
138 import shap
139 import numpy as np
140 import matplotlib.pyplot as plt
141
142 # Contexte
143 churn_csv = r"C:\Users\mehdi\Documents\Mehdi\HEC\Cours\Python\Project\
       Telco_Customer_Churn.csv"
144
145 df = pd.read_csv(
146     churn_csv,
147     sep=',',
148     engine='python',
149     on_bad_lines='skip',
150     encoding='utf-8'
151 )
152
153 df['Churn'] = df['Churn'].map({'Yes': 1, 'No': 0})
154 df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
155 df = df.dropna()
156
157 df = df.drop(columns = ['customerID'])
158
159 df['gender'] = df['gender'].map({'Female': 1, 'Male': 0})
160 df['Partner'] = df['Partner'].map({'Yes': 1, 'No': 0})
161 df['Dependents'] = df['Dependents'].map({'Yes': 1, 'No': 0})
162 df['PhoneService'] = df['PhoneService'].map({'Yes': 1, 'No': 0})
163 df['MultipleLines'] = df['MultipleLines'].map({'Yes': 1, 'No': 0, 'No
       phone service' : 0.5})
164 df['InternetService'] = df['InternetService'].map({'Fiber optic': 1, 'No
       ': 0, 'DSL' : 0.5})
165 df['OnlineSecurity'] = df['OnlineSecurity'].map({'Yes': 1, 'No': 0, 'No
       internet service' : 0.5})
166 df['OnlineBackup'] = df['OnlineBackup'].map({'Yes': 1, 'No': 0, 'No
       internet service' : 0.5})
167 df['DeviceProtection'] = df['DeviceProtection'].map({'Yes': 1, 'No': 0,
       'No internet service' : 0.5})
168 df['TechSupport'] = df['TechSupport'].map({'Yes': 1, 'No': 0, 'No
       internet service' : 0.5})
```

```
df['StreamingTV'] = df['StreamingTV'].map({'Yes': 1, 'No': 0, 'No internet service' : 0.5})
df['StreamingMovies'] = df['StreamingMovies'].map({'Yes': 1, 'No': 0, 'No internet service' : 0.5})
df['Contract'] = df['Contract'].map({'Month-to-month': 1, 'One year': 12, 'Two year' : 24})
df['PaperlessBilling'] = df['PaperlessBilling'].map({'Yes': 1, 'No': 0})
df['PaymentMethod'] = df['PaymentMethod'].map({'Electronic check': 0.5, 'Mailed check' : 0.25, 'Bank transfer (automatic)': 0.75,'Credit card (automatic)': 1})
print(df.describe())

print(df.head())

# ML

Y = df['Churn']
features = ['gender', 'Partner', 'Dependents', 'PhoneService', 'MultipleLines',
            'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
            'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract',
            'PaperlessBilling', 'PaymentMethod']

X = df[features]


X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state = 42, stratify = Y)

model = RandomForestClassifier(n_estimators = 300, random_state = 42, class_weight = 'balanced')
model.fit(X_train, Y_train)

Y_pred = model.predict(X_test)
Y_prob = model.predict_proba(X_test)[:,1]

print("Accuracy:", accuracy_score(Y_test, Y_pred))
print("Classification Report:", classification_report(Y_test, Y_pred))
auc = roc_auc_score(Y_test, Y_prob)
print("ROC AUC:", auc)

fpr, tpr, _ = roc_curve(Y_test, Y_prob)

plt.figure(figsize=(6,6))
plt.plot(fpr, tpr, label=f"AUC = {auc:.3f}")
plt.plot([0,1],[0,1],'--')
```

```
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - Churn Prediction")
plt.legend()
plt.show()

# Precision Recall Curve

p,r,_ = precision_recall_curve(Y_test, Y_prob)

plt.figure()
plt.plot(r,p)
plt.xlabel("Recall")
plt.ylabel("Precision")
plt.title("Precision Recall Curve")
plt.show()

importances = pd.Series(model.feature_importances_, index=X.columns)\
                .sort_values(ascending=False)


# Shap Explainability

# use small sample for speed
X_shap = X_test.sample(300, random_state=42)

# Explainer pour RandomForestClassifier
explainer = shap.Explainer(model, X_train, model_output="probability")

shap_values = explainer(X_shap)

# Beeswarm plot
shap.plots.beeswarm(shap_values)

# Bar plot (global importance)
shap.plots.bar(shap_values)


plt.figure(figsize=(8,6))
importances.head(10).plot(kind='barh')
plt.title("Top 10 Feature Importance")
plt.gca().invert_yaxis()
plt.show()

print("\nTop features:\n", importances.head(10))


```

```
# Correlation Heatmap
plt.figure(figsize=(12,8))
sns.heatmap(df.corr(numeric_only=True), annot=True, fmt = ".2f")
plt.title("Correlation Heatmap")
plt.show()
```

A.3 XGBoost

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import shap

from xgboost import XGBClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import (
    classification_report,
    roc_auc_score,
    roc_curve,
    precision_recall_curve,
    accuracy_score
)

# Contexte

churn_csv = r"C:\Users\mehdi\Documents\Mehdi\HEC\Cours\Python\Project\
    Telco_Customer_Churn.csv"

df = pd.read_csv(churn_csv)

df['Churn'] = df['Churn'].map({'Yes':1,'No':0})

df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df = df.dropna()

df = df.drop(columns=['customerID'])

binary_map = {'Yes':1,'No':0}

df['gender'] = df['gender'].map({'Female':1,'Male':0})
df['Partner'] = df['Partner'].map(binary_map)
df['Dependents'] = df['Dependents'].map(binary_map)
df['PhoneService'] = df['PhoneService'].map(binary_map)
df['PaperlessBilling'] = df['PaperlessBilling'].map(binary_map)

service_map = {'Yes':1,'No':0,'No internet service':0.5}
df['OnlineSecurity'] = df['OnlineSecurity'].map(service_map)
df['OnlineBackup'] = df['OnlineBackup'].map(service_map)
df['DeviceProtection'] = df['DeviceProtection'].map(service_map)
df['TechSupport'] = df['TechSupport'].map(service_map)
df['StreamingTV'] = df['StreamingTV'].map(service_map)
df['StreamingMovies'] = df['StreamingMovies'].map(service_map)

```

```
df['MultipleLines'] = df['MultipleLines'].map({'Yes':1,'No':0,'No phone service':0.5})
df['InternetService'] = df['InternetService'].map({'Fiber optic':1,'DSL':0.5,'No':0})
df['Contract'] = df['Contract'].map({'Month-to-month':0,'One year':1,'Two year':2})
df['PaymentMethod'] = df['PaymentMethod'].map({
    'Electronic check':0,
    'Mailed check':0.3,
    'Bank transfer (automatic)':0.7,
    'Credit card (automatic)':1
})

# ML

features = [
    'tenure','MonthlyCharges','TotalCharges',
    'Contract','OnlineSecurity','TechSupport',
    'InternetService','PaymentMethod',
    'PaperlessBilling','SeniorCitizen'
]

X = df[features]
y = df['Churn']



X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    stratify=y,
    random_state=42
)

background = shap.sample(X_train, 100)


model = XGBClassifier(
    n_estimators=400,
    max_depth=4,
    learning_rate=0.05,
    subsample=0.8,
    colsample_bytree=0.8,
    scale_pos_weight=3,
    eval_metric='logloss',
    use_label_encoder=False,
    base_score=0.5
```

```
)

model.fit(X_train, y_train)



Y_prob = model.predict_proba(X_test)[:,1]

threshold = 0.35
preds = (Y_prob > threshold).astype(int)



print("\nAccuracy:", accuracy_score(y_test, preds))
print("\nClassification Report:\n", classification_report(y_test, preds)
    )

auc = roc_auc_score(y_test, Y_prob)
print("ROC AUC:", auc)


# Roc Curve

fpr,tpr,_ = roc_curve(y_test, Y_prob)

plt.figure(figsize=(6,6))
plt.plot(fpr, tpr, label=f"AUC = {auc:.3f}")
plt.plot([0,1],[0,1],'--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - Churn Prediction")
plt.legend()
plt.show()


# Precision Recall Curve

p,r,_ = precision_recall_curve(y_test, Y_prob)

plt.figure()
plt.plot(r,p)
plt.xlabel("Recall")
plt.ylabel("Precision")
plt.title("Precision Recall Curve")
plt.show()


```

```
# Feature Importance

imp = pd.Series(model.feature_importances_, index=features).sort_values
    ()

plt.figure(figsize=(8,6))
imp.head(10).plot(kind='barh')
plt.title("Top 10 Feature Importance")
plt.gca().invert_yaxis()
plt.show()

print("\nTop features:\n", imp.head(10))


# Shap Explainability
explainer = shap.TreeExplainer(model.get_booster())
shap_values = explainer.shap_values(X_test)

shap.summary_plot(shap_values, X_test, feature_names=X_test.columns)
shap.summary_plot(shap_values, X_test, feature_names=X_test.columns,
    plot_type="bar")

# Business List - Top Churn Risk Customers

risk = X_test.copy()
risk['Churn_Probability'] = Y_prob
risk = risk.sort_values('Churn_Probability', ascending=False)

print("\nTop 10 Customers at Risk:\n")
print(risk.head(10))
```

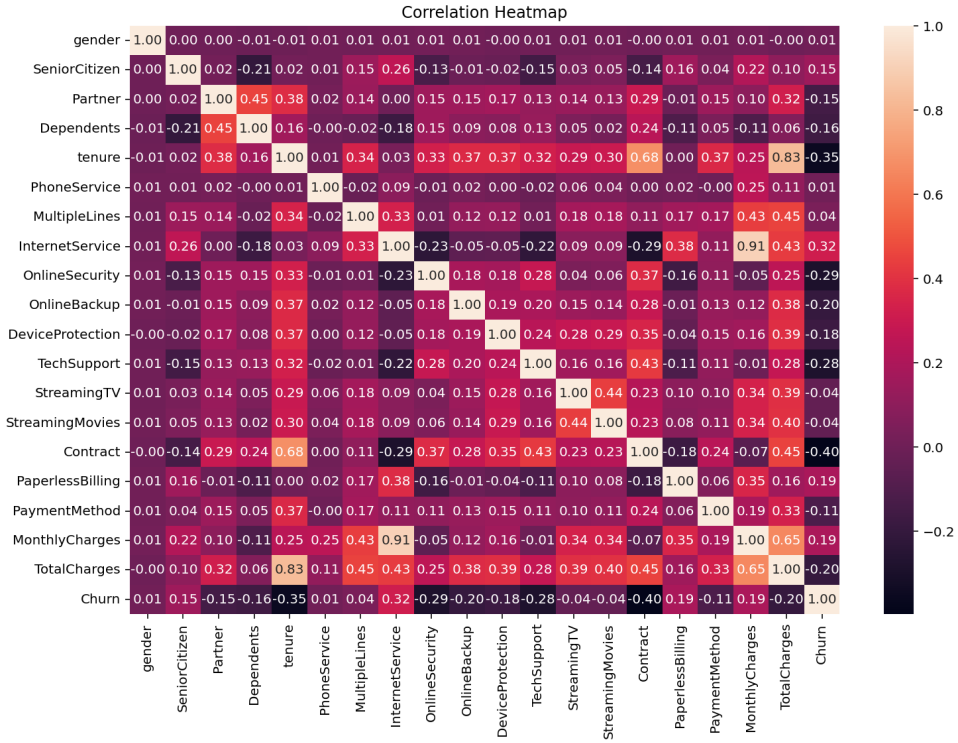


Figure A.1: Correlation Heatmap

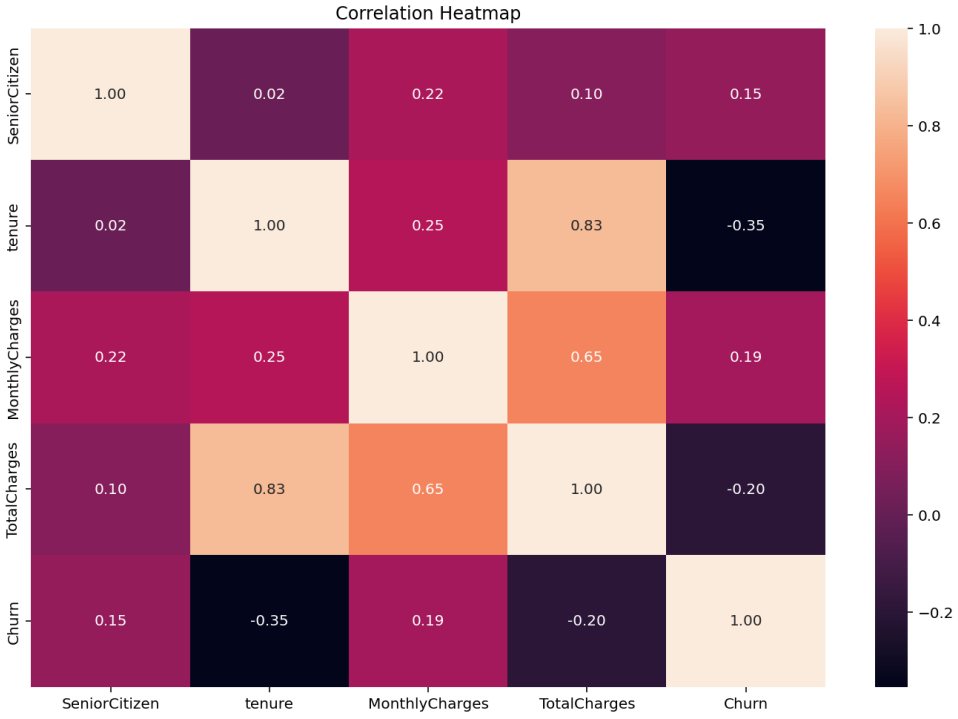


Figure A.2: Correlation Heatmap - 5 Features

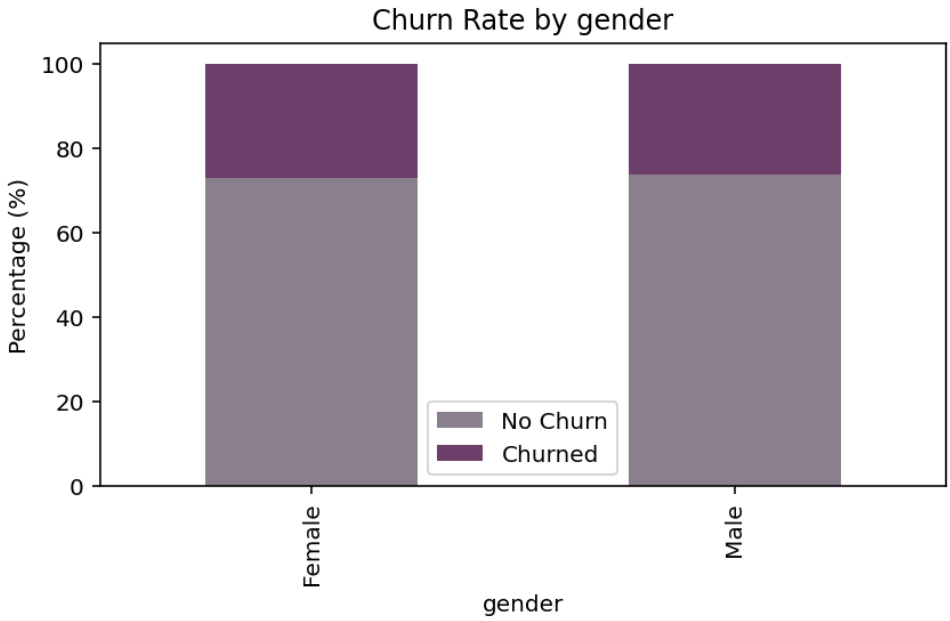


Figure A.3: Churn Rate by Gender

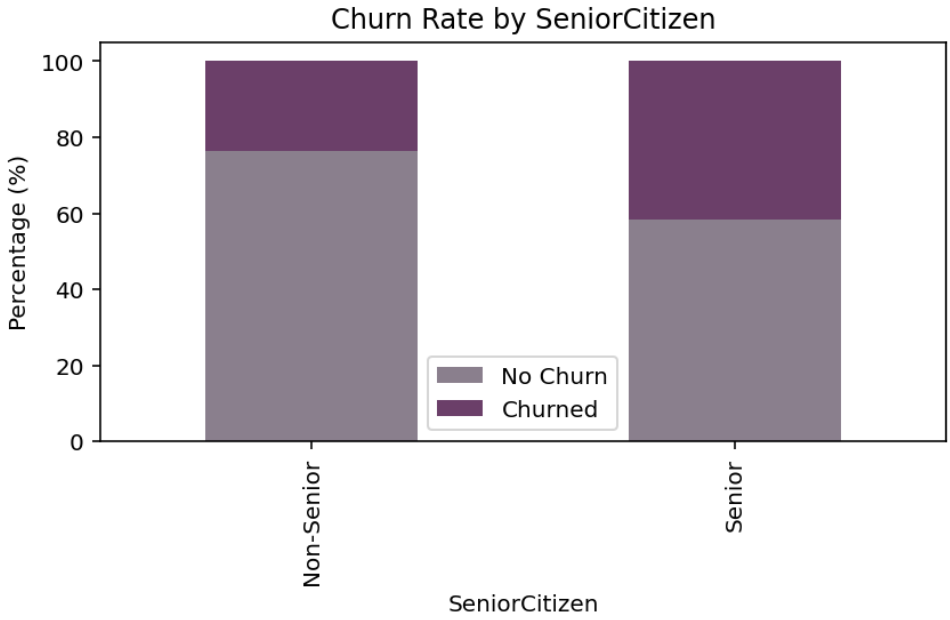


Figure A.4: Churn Rate by Seniority

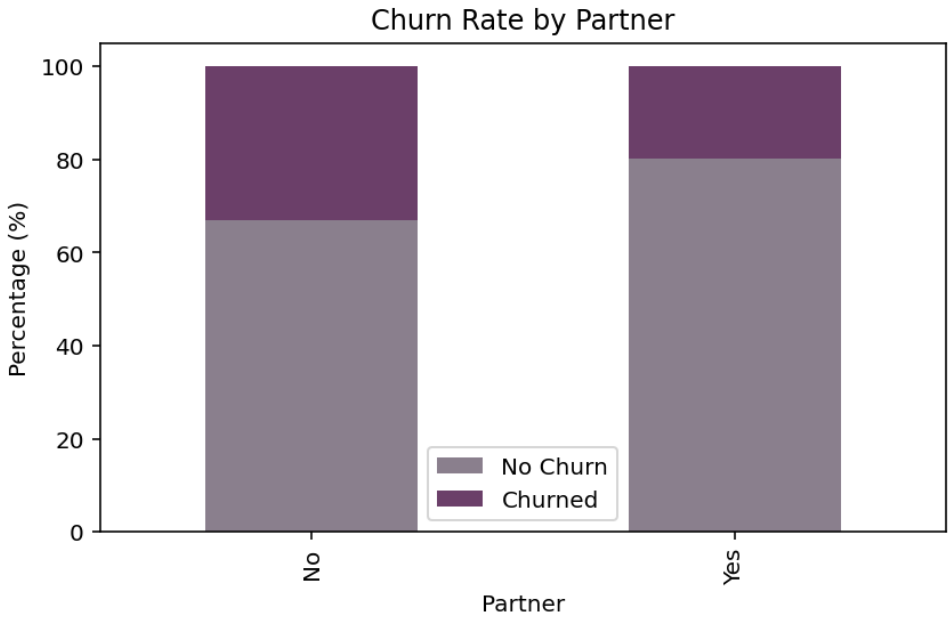


Figure A.5: Churn Rate by Partnership

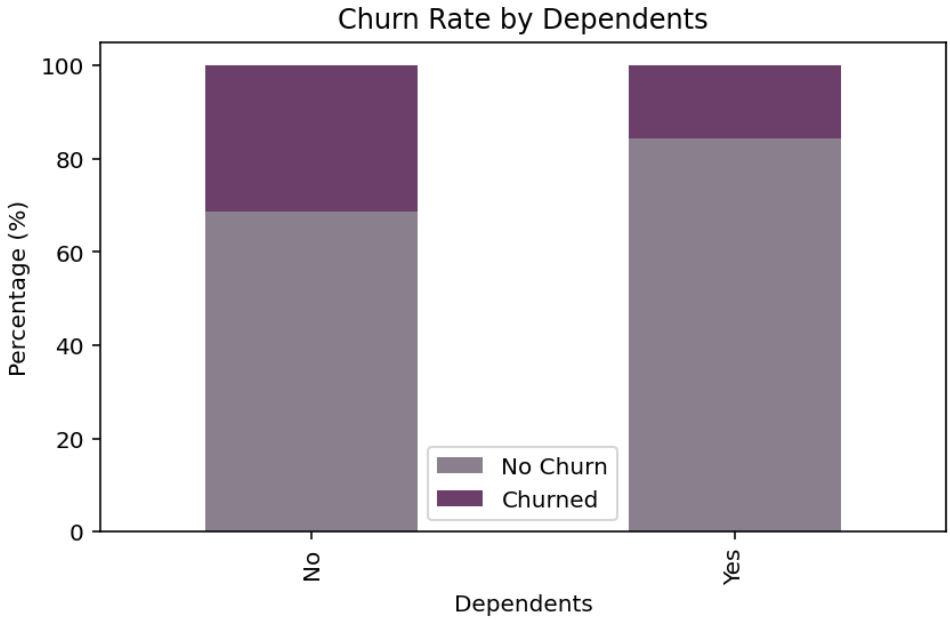


Figure A.6: Churn Rate by Dependency